

Disk heating due to planet-induced spiral shocks and implications to a snowline

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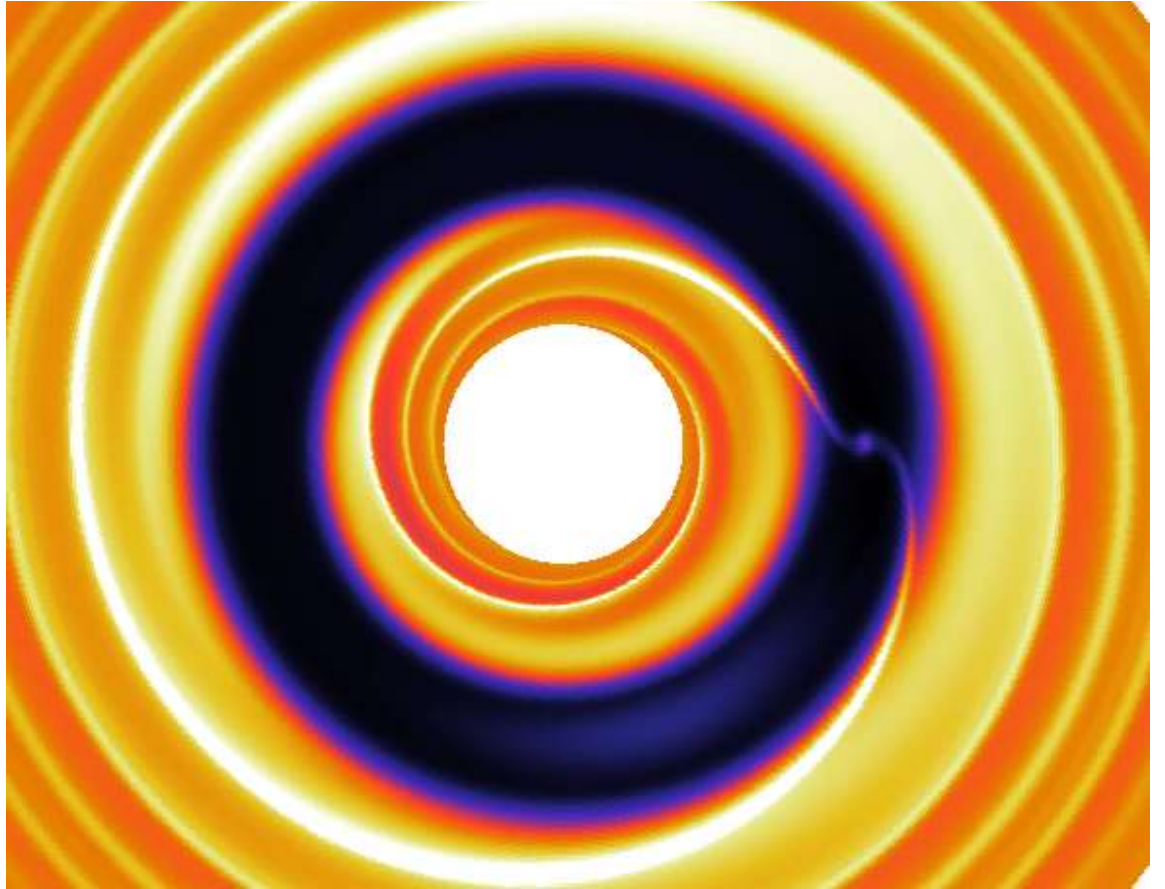
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Disk-Planet Interaction

- Gap+spiral structures in a protoplanetary disk
- Focus on spiral wave propagation and its consequences



Courtesy: K. Kanagawa

Spiral Wave Propagation in a Disk

- Spiral wave = Sound wave propagating radially away from the planet
- “Characteristic curve” of 2D supersonic steady flow
 - Spiral shape
 - Distance of the shock formation

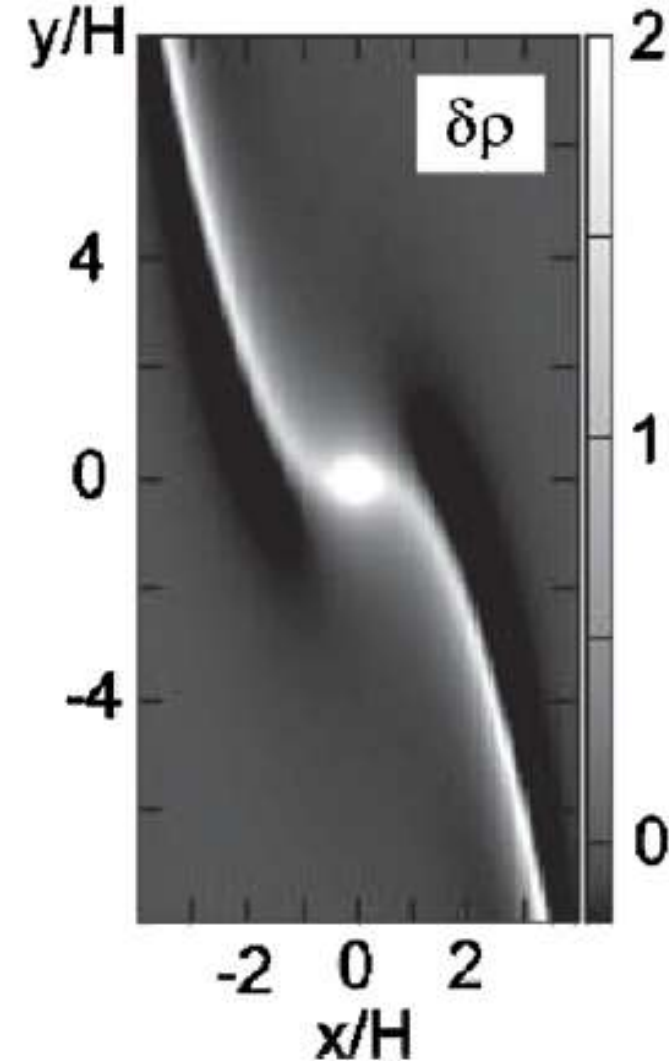
$$\left(\frac{dy}{dx}\right)_{\pm} = \frac{v_x v_y \pm c \sqrt{v^2 - c^2}}{v_x^2 - c^2}.$$

Shape of the main spiral:

$$y \sim -\frac{3}{4} \frac{x^2}{H} + y_0,$$

Distortion:

$$\delta y(x) \propto M_p x^{5/2} g(y_0),$$



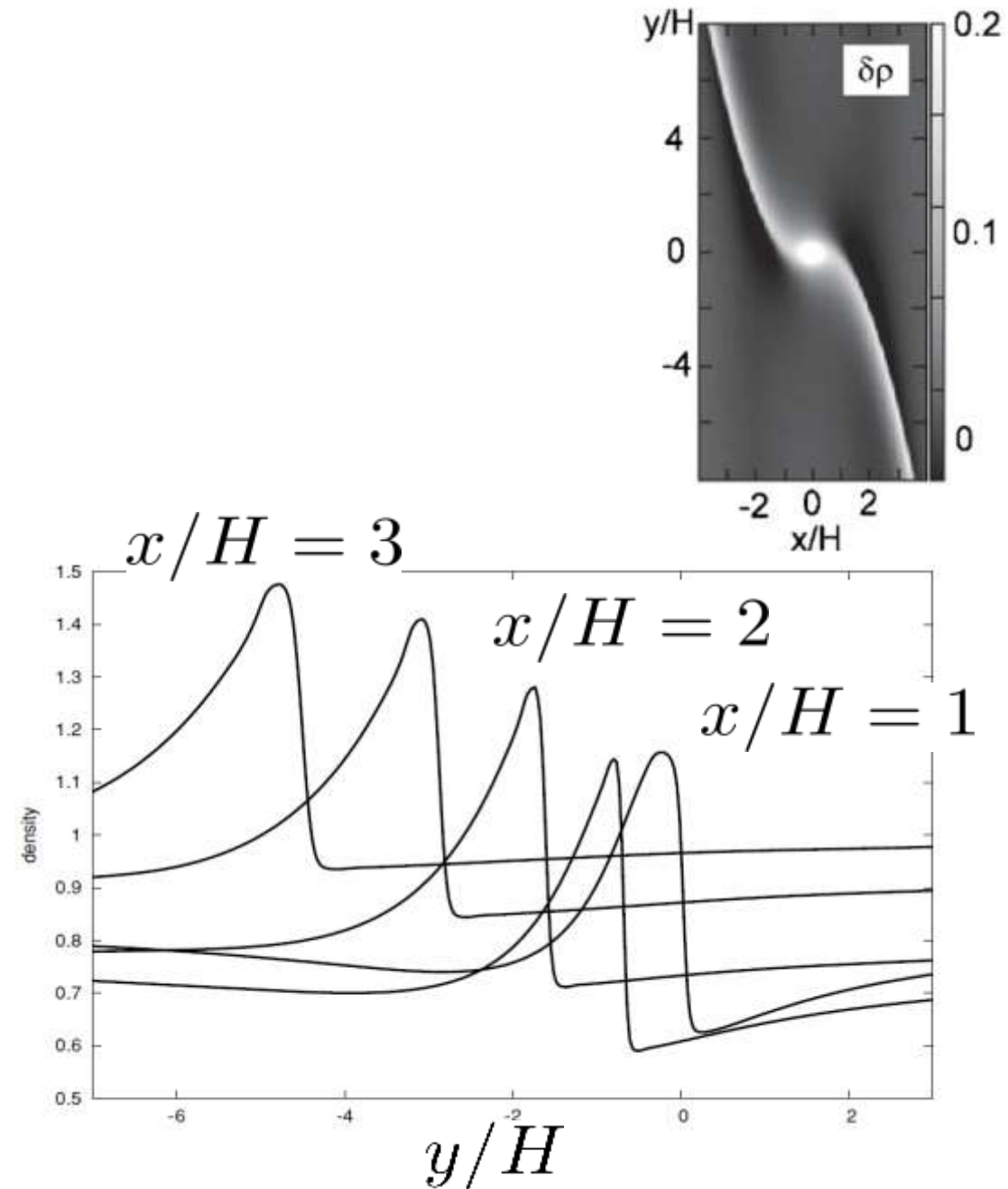
Spiral Wave Shocks

- Sound wave shocks as it propagates [Goodman and Rafikov 2001]
- Shock distance [Dong et al. 2011]

$$x_s = 0.8 \left(\frac{\gamma + 1}{12/5} \frac{M_p}{M_{th}} \right)^{-2/5} H,$$

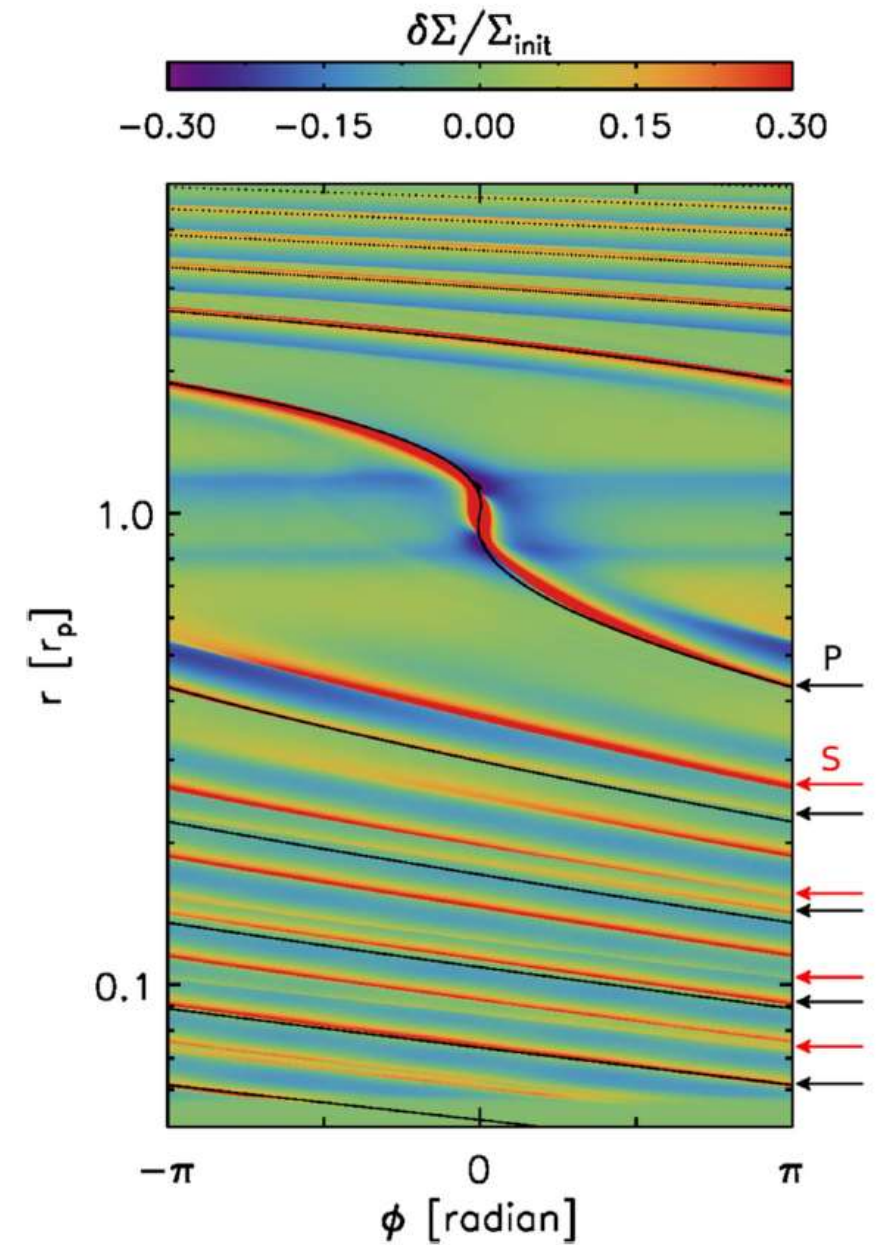
Maybe understood by the distortion of the characteristic curve

$$\delta y(x) \propto M_p x^{5/2} g(y_0),$$



Multiple Spiral Waves

- A planet induces multiple waves in a global disk

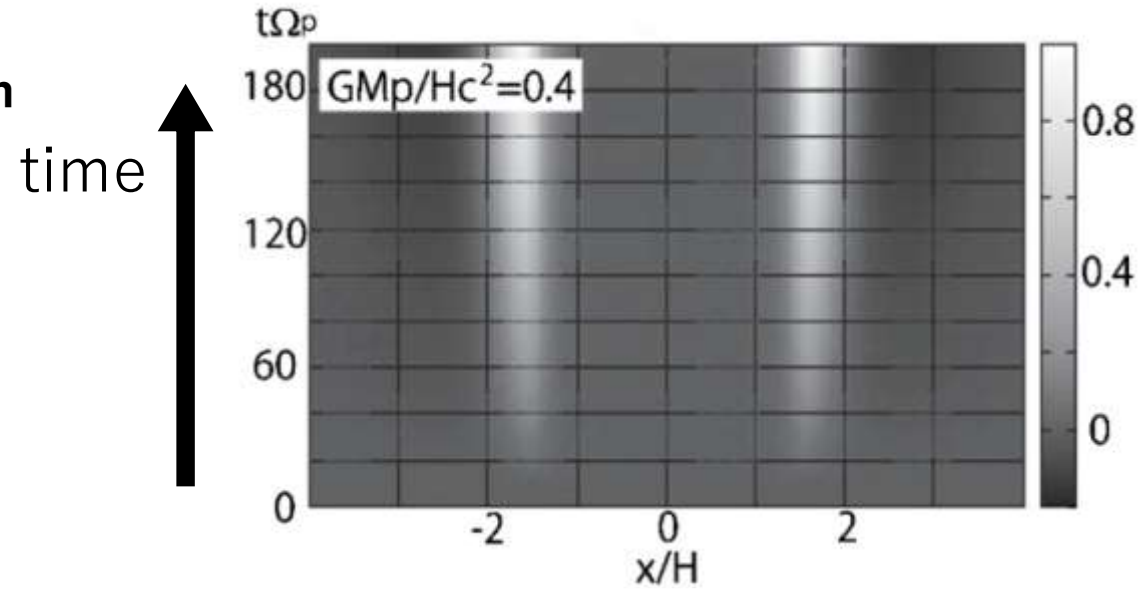


Consequences of Shock Formation

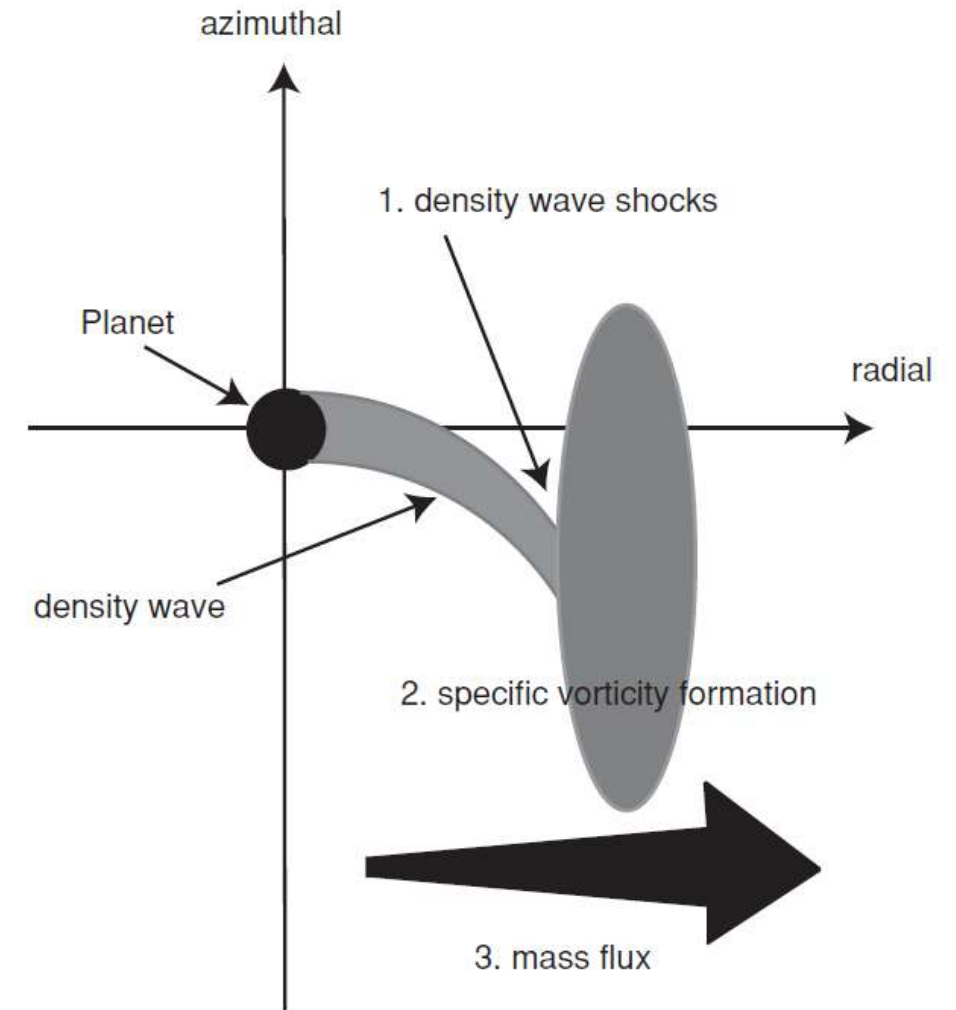
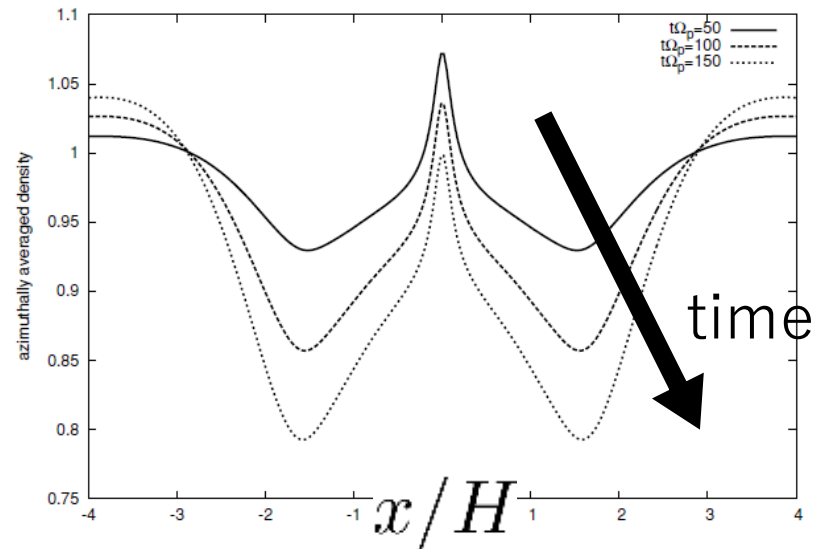
- **Vortex formation and gap formation (for a primary wave)**
- Heating of the disk
- Gap-Heating Connection

Vortex Formation by Spiral Shock

Vorticity
evolution



Density
evolution



Gap Formation Process

- Vortex acts as a “source” of radial mass flux
- ... And the radial force balance should be properly treated

$$R \frac{\partial \Sigma}{\partial t} + \frac{\partial}{\partial R} (R \Sigma v_R) = 0,$$

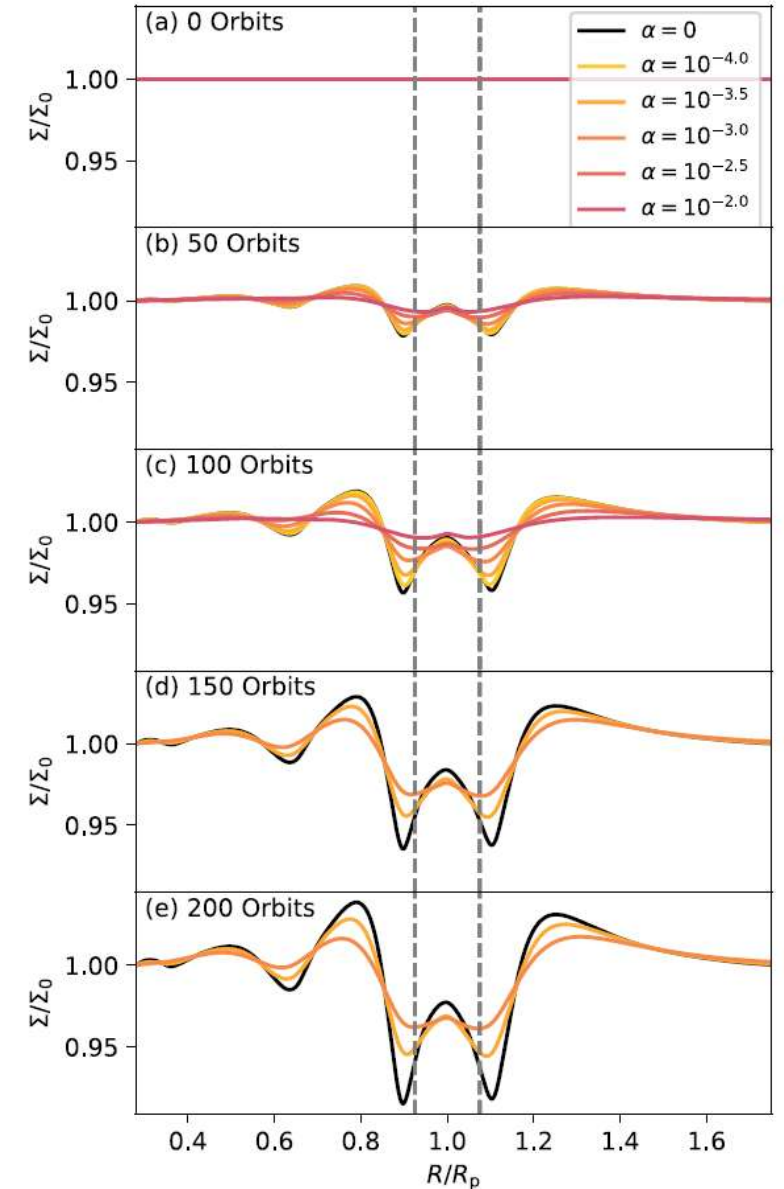
Viscosity

Vorticity

$$R \frac{\partial}{\partial t} (\Sigma R^2 \Omega) + \frac{\partial}{\partial R} (R \Sigma v_R \Omega R^2) = -\frac{1}{2\pi} \left(\frac{\partial G}{\partial R} - \frac{\partial F_{\text{dep}}}{\partial R} \right),$$

$$\Omega^2 = \Omega_K^2 + \frac{1}{R \Sigma} \frac{\partial P}{\partial R},$$

- **Gap depth evolves linearly with time in the case of an inviscid disk** [Cordwell & Rafikov 2024]
- **Vortex production due to the shock is the (only) model needed** [Cimerman & Rafikov 2023]

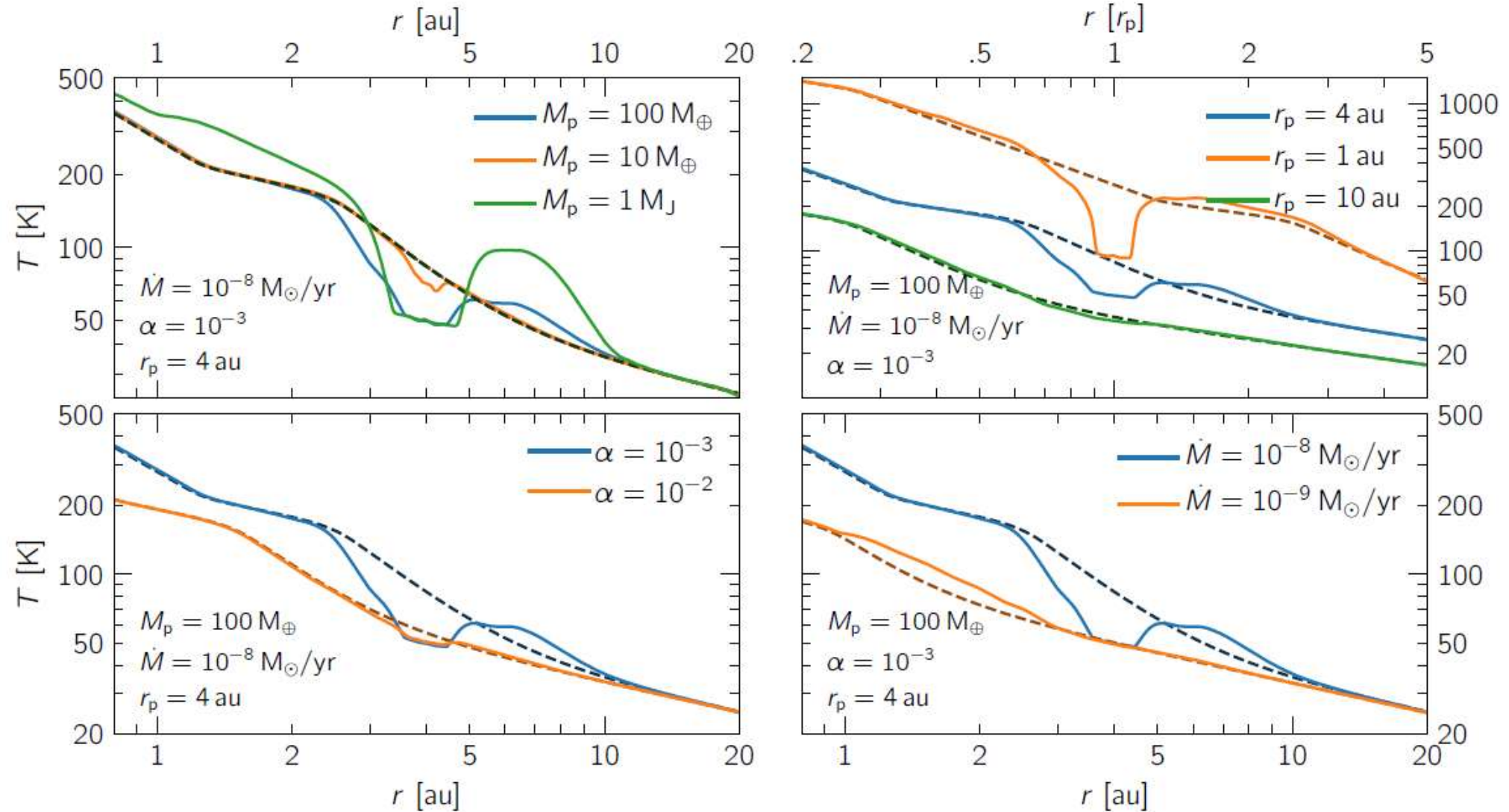


Cordwell & Rafikov (2024)

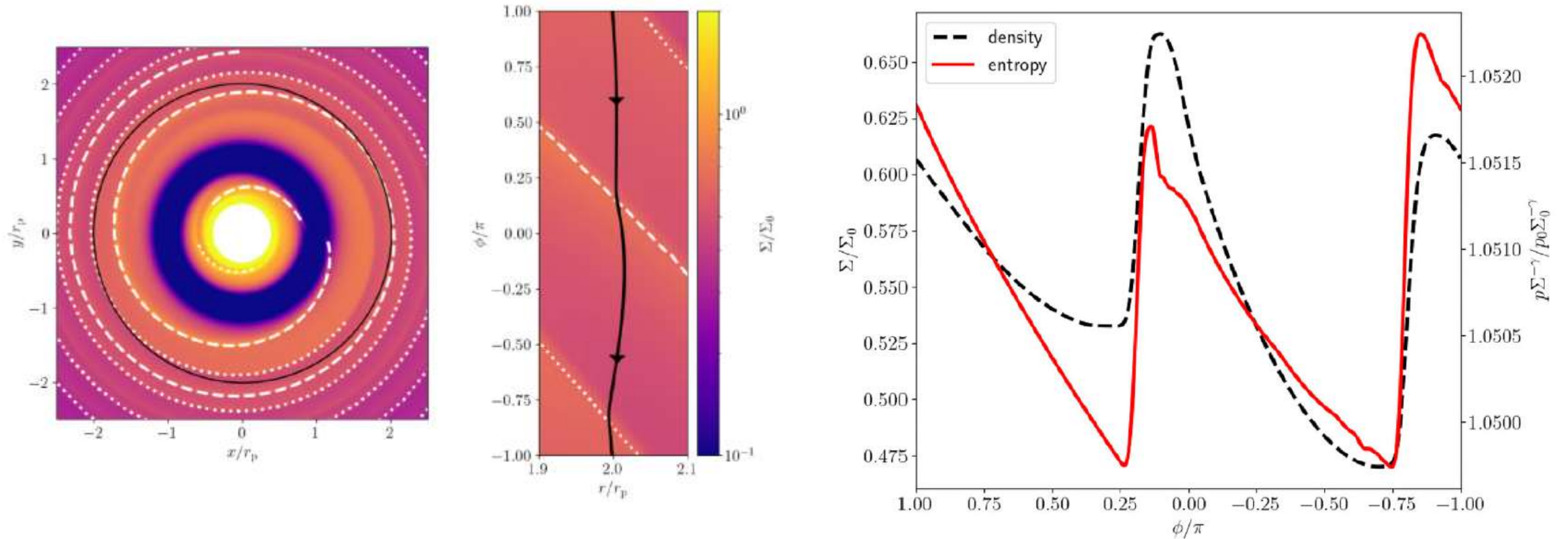
Consequences of Shock Formation

- Vortex formation and gap formation
- **Heating of the disk**
- Gap-Heating Connection

A Planet Heats the Disk



Spiral Shock and Streamline



- Entropy jump at the spiral shock acts as a source of heating

$$\delta \equiv \frac{s_2}{s_1} - 1 = \frac{p_2/\Sigma_2^\gamma}{p_1/\Sigma_1^\gamma} - 1.$$

$$Q_{\text{sh}} = \frac{\Sigma c_s^2}{\Delta t(\gamma - 1)} \delta = \frac{\Sigma e}{2\pi} |\Omega_K - \Omega_p| \delta,$$

Empirical Fit of Entropy Jump

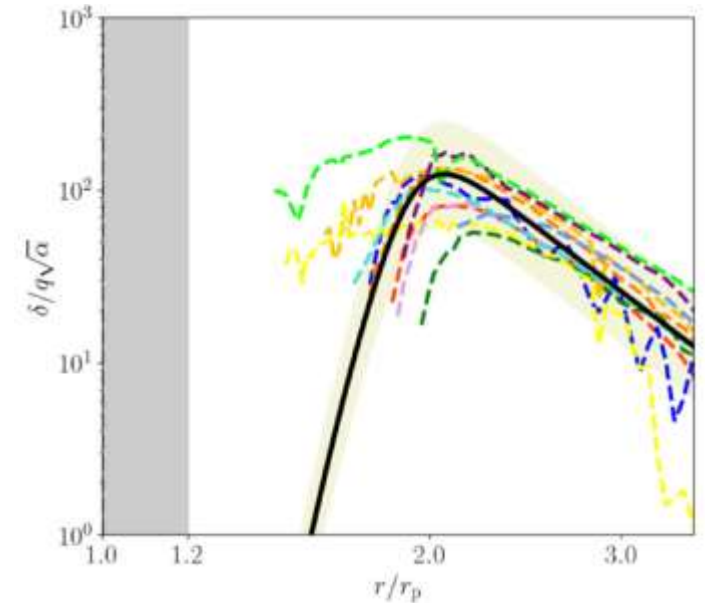
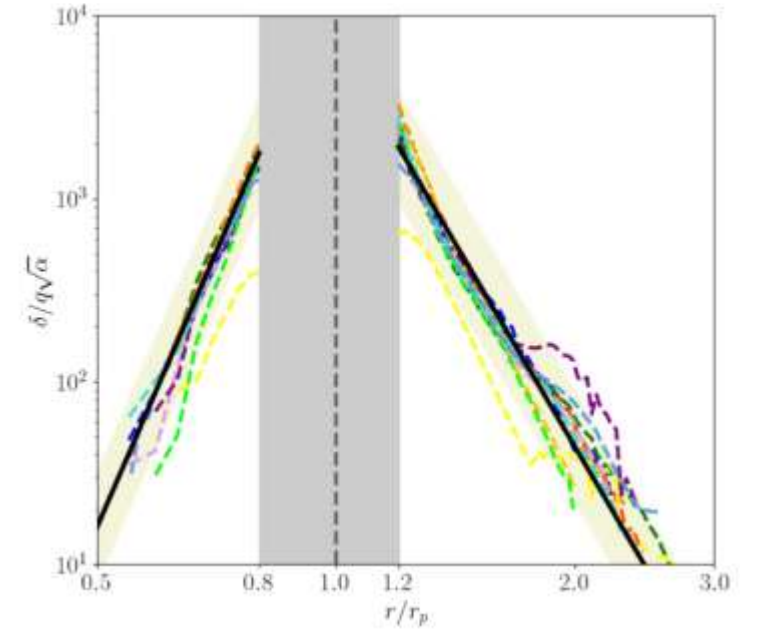
Primary: $\delta = 10^{f_0} q \sqrt{\alpha} \cdot |r/r_p|^{f_1}$

Outer Secondary: $\delta = 10^{3.7} q \sqrt{\alpha} \left(\frac{r}{r_p}\right)^{-4.79} \left[\left(\frac{r/r_p - 1}{0.95}\right)^{-12} + 1 \right]^{-1}$

- Derived from simulations with beta-cooling prescription and

$$10^{-4} \leq \frac{M_p}{M_*} \leq 10^{-3} \quad 10^{-4} \leq \alpha \leq 10^{-3}$$

$$10 \leq \beta \leq 10^{2.5}$$



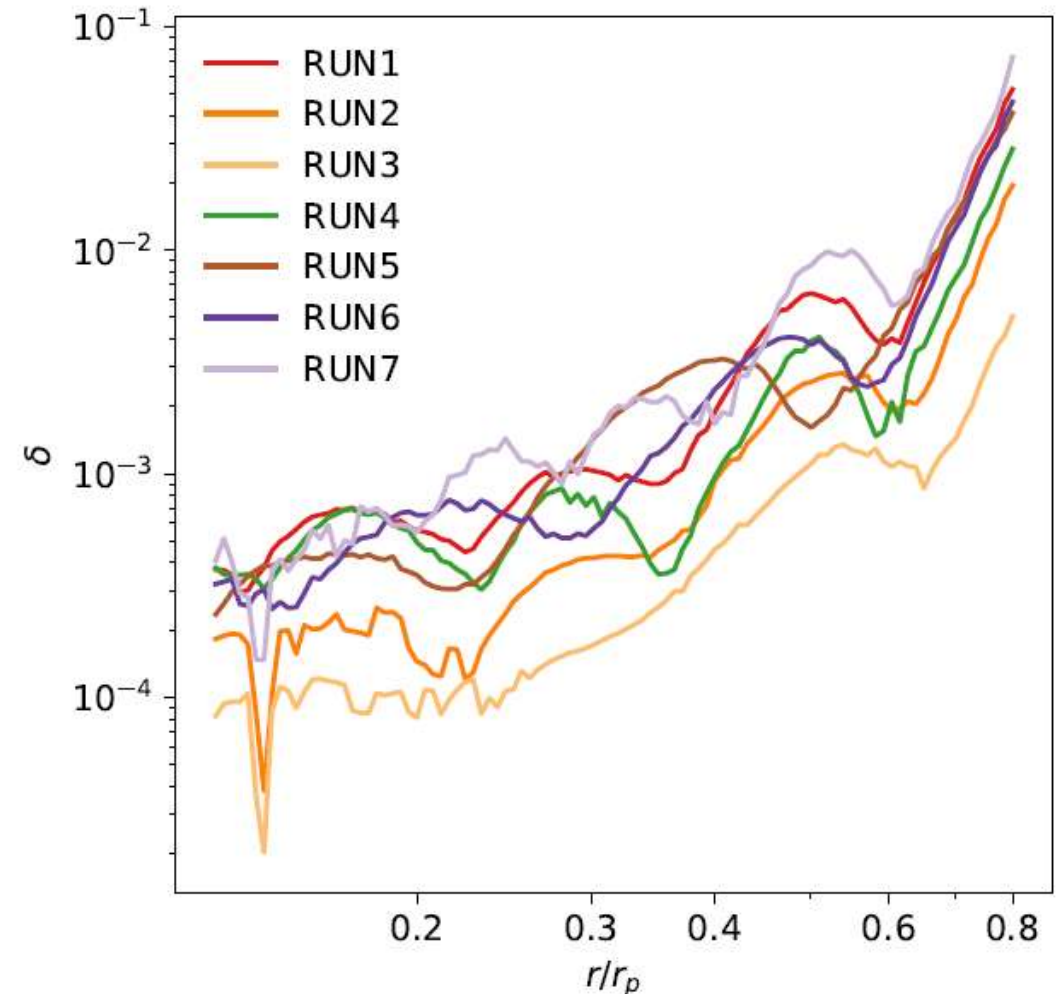
And with More Extended Inner Boundary

- The inner boundary of the simulations are extended to include higher-order spirals

$$10^{-4} \leq \frac{M_p}{M_*} \leq 10^{-3}$$

$$10^{-3.5} \leq \alpha \leq 10^{-3}$$

$$\beta = 100$$



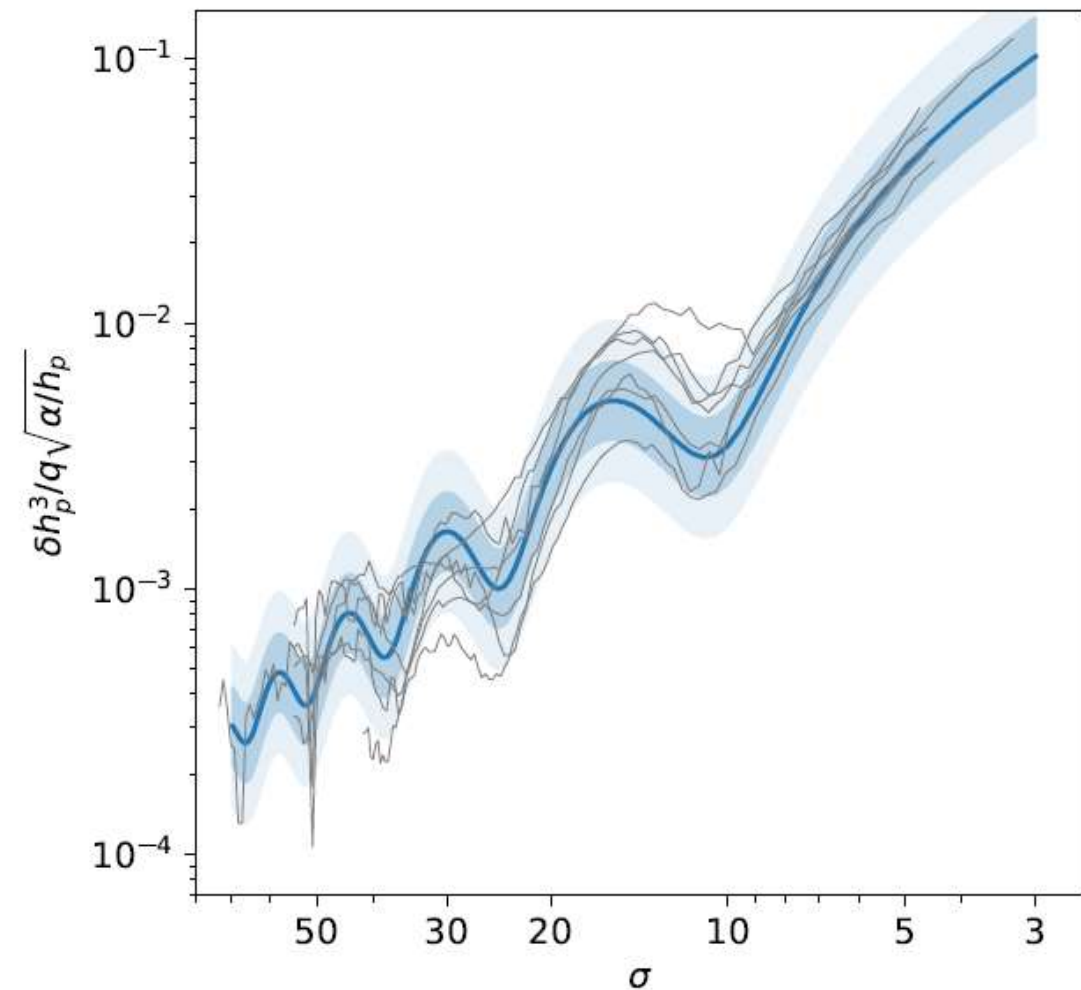
Shimizu et al. in prep.

Entropy Jump and Scaling

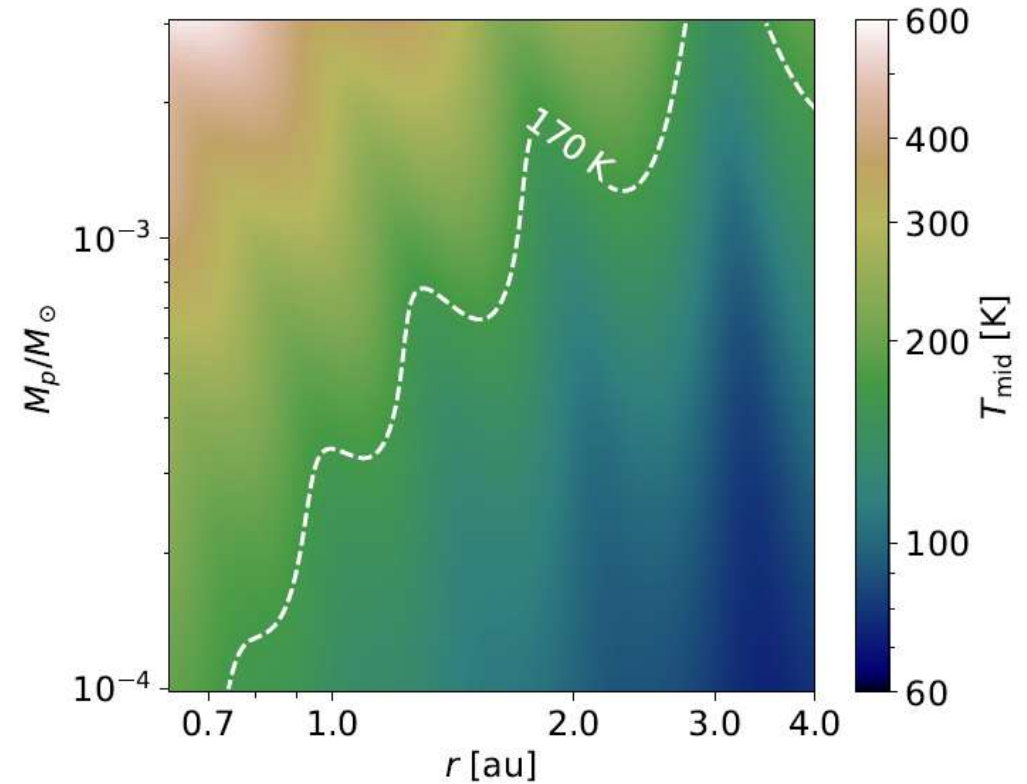
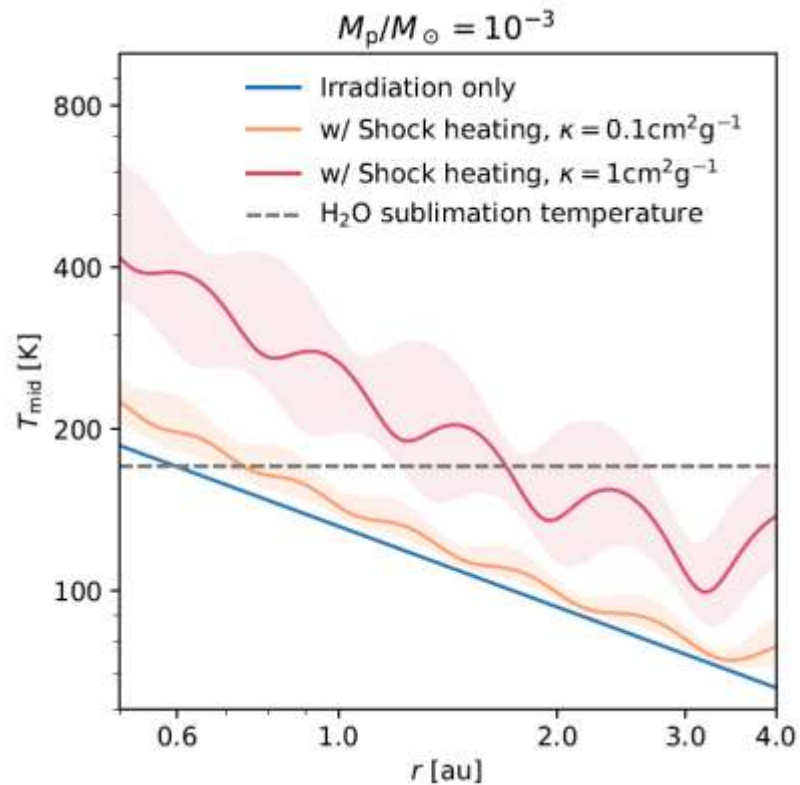
- Radial distance scaled by local scale height
- “Wavy” function to account for heating by multiple spirals

$$\sigma = \int_r^{r_p} \frac{dr}{H(r)}$$

$$\delta = \mathcal{A} \sqrt{\frac{\alpha}{h_p} \frac{M_p}{M_{\text{th}}}} \sigma^{\mathcal{B}} (\mathcal{C} \exp(-\mathcal{D}\sigma) \sin(\sigma/\mathcal{E}) + 1)$$



Snow Line Location



- A Jupiter at 5au can allow a rocky (non-icy) environment at 1au

Consequences of Shock Formation

- Vortex formation and gap formation
- Heating of the disk
- **Gap-Heating Connection**

Gap-Heating Connection

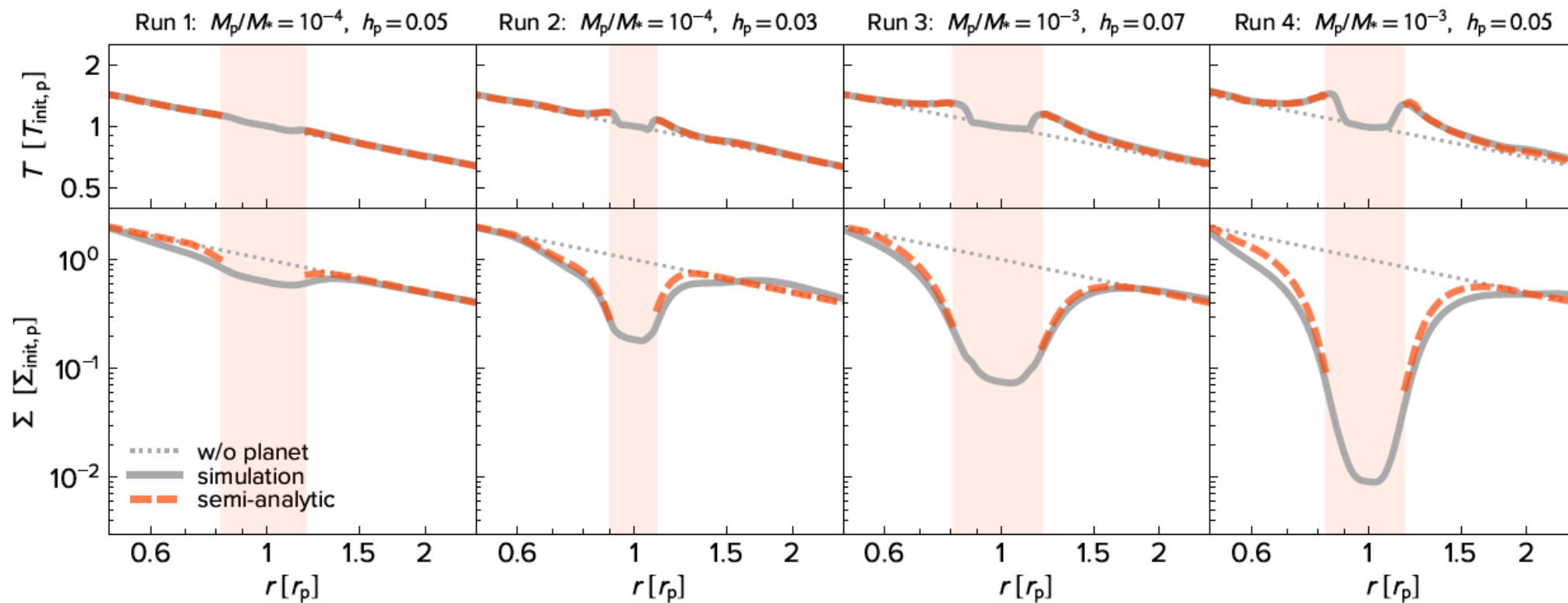
- Angular momentum transfer and the disk heating are both due to the spiral shock

$$\Lambda_{\text{shock}}(r) = \frac{2\pi r Q_{\text{shock}}(r)}{\Omega_{\text{p}} - \Omega_{\text{K}}(r)}$$

- Gap structure and disk temperature can be calculated consistently

$$\frac{\partial \ln(r^2 \alpha c_T^2 \Sigma)}{\partial \ln r} = \frac{\sigma \delta}{3\pi(\gamma - 1)\alpha} \quad T = T_{\text{init}} \left(1 - \frac{\beta |1 - \Omega_{\text{p}}/\Omega_{\text{K}}|}{2\pi} f_{\text{heat}} \delta \right)^{-1}$$

Gap-Heating Connection



Summary

- Spiral density wave launched by a planet affects disk physical structures
- Gap formation in a disk by angular momentum transfer
- Disk heating associated with entropy production
- Gap structure and disk heating are connected – Physics controlled primarily by the planet mass